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Hybrid Multi-Agent Systems for Auditable AI Surveying

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Abstract

Systematically acquiring expert knowledge remains a bottleneck. Large Language Models (LLMs) scale interaction but introduce a governance challenge: inconsistent coverage, topic drift, and user steering. We present *MHAESTRO*, a hybrid two-phase approach that aims for both scale and accountable control. In Phase 1, a Knowledge-Engineering tool (*K-Eng*) compiles expert input into a versioned decision tree. In Phase 2, a Multi-Agent Elicitation tool (*Elicitor*) conducts a tightly structured conversational survey that strictly traverses this deterministic policy while using LLMs for phrasing and summarisation. Internal structured control ensures the user is never the last point of control, reducing steering and aligning runs to the mandated structure. We report a formative case study (N=8) that assessed extraction efficacy and user experience. Extraction fidelity was high: 75% agreed the end-of-session summary accurately captured their input. Ease of use was also high (87.5%). However, participants reported high perceived intrusiveness, evidencing a fidelity–fluidity trade-off whereby governance mechanisms that enforce coverage can increase interactional strain. We argue this is not merely a design issue but a material accessibility and equity concern, as such strain may disproportionately affect people with high cognitive fatigue or neurodivergence. Our findings show that deterministic control can be combined with LLM generation to deliver rigorous, auditable surveying, but at a human-centred cost that must be actively managed. We outline design and governance implications for accessible, equitable AI-mediated conversational surveying and note the architectural potential for real-time safety monitoring agents.

Keywords: Dialogue policy, AI governance, Auditability, Knowledge elicitation, Knowledge engineering, Multi-Agent system

1 Introduction

Institutions increasingly need surveys and interview-style engagements that scale like surveys while remaining accountable this includes consistent coverage, explainable routing, auditable artefacts, and safeguards for participant comfort and safety (Groves et al. 2009; Mitchell et al. 2019; Raji et al. 2020). Getting this right matters beyond system design. Public-facing surveys shape policy and services; if governed AI increases burden or deters participants, it risks excluding voices and reducing the legitimacy

of the evidence base. Free-form LLM conversational surveying improves phrasing but risks drift, inconsistent coverage and poor traceability (Ji et al. 2023; West et al. 2018; Schober and Conrad 1997). We address this by separating the conversational survey policy (what must be asked, in what order, and when to stop) from its execution at scale. The policy is a versioned decision graph; execution is a governed flow that blocks progression until adequacy is met and logs the basis for movement.

This challenge, however, is not purely technical; it is socio-technical (Mittelstadt et al. 2016; Jobin et al. 2019). As AI-mediated systems become primary instruments for social science and knowledge elicitation, we face a critical tension: how to enforce the scientific rigour of structured elicitation without imposing interactional strain that dehumanises participants and risks invalidating the very knowledge we seek to acquire (Tourangeau et al. 2000; Schober and Conrad 1997; West et al. 2018; Galesic and Bosnjak 2009). This paper investigates this tension, arguing that the design of AI governance has direct, measurable, and ethically significant consequences for human experience, particularly for accessibility and equity (Mitchell et al. 2019; Gebru et al. 2021; Raji et al. 2020; Lazar et al. 2015).

We ask how can AI systems deliver auditable control in conversational knowledge elicitation without undermining human agency and accessibility? Our contributions are: (a) a two-phase sequential architecture that separates knowledge engineering from execution and encodes the conversational survey rules as a a predefined, rule-based question sequence (encoded as JSON); (b) a multi-agent elicitation pipeline in which traversal and adequacy decisions are carried as internal control messages, so progression cannot be misdirected by off-topic user turns, mitigating user steering; and (c) a formative evaluation that empirically identifies and quantifies a core fidelity-fluidity trade-off, revealing the human-centric cost of governance-by-design and its critical implications for accessibility and validity.

1.1 The Challenge of Expert Knowledge Acquisition

Despite significant advances in the field of Artificial Intelligence, The "knowledge acquisition bottleneck" remains a pivotal challenge in the development of intelligent systems. Eliciting expert knowledge is demanding, context-sensitive, and labour-intensive, often relying on scarce human specialists and repeated rounds of conversational surveying and analysis (Hoffman 1989; Shadbolt and Smart 2015). This limits scale, consistency, and auditability. For settings that resemble large surveys where many respondents are asked the same structured questions with reliable coverage the usual ad-hoc conversational surveys do not provide the needed uniformity or a clear trace from prompts to recorded knowledge (Tourangeau et al. 2000; Groves et al. 2009; Raji et al. 2020). A practical route forward is to separate the planning of a conversational survey (what must be asked, in what order, and under which conditions) from the controlled execution of that plan with many respondents.

1.2 Limitations of Generative models in Structured Elicitation

LLMs are strong at fluent dialogue and, with instruction-following training, can adapt phrasing to user needs (e.g., RLHF-trained models) (Ouyang et al. 2022; DeTemple and Meine 2025). However, they operate probabilistically and may vary content between runs, which creates a risk for structured elicitation where the exact coverage and path matter (Ji et al. 2023; Mitchell et al. 2019; Gebru et al. 2021; Carlini et al. 2021). In expert-facing contexts this variability weakens repeatability, blurs responsibility, and complicates auditing. What is needed is a method to keep the generative benefits (flexible phrasing, summarisation, schema filling) while constraining the control flow so that coverage is enforced and each step is accountable.

1.3 Our approach: hybrid control for survey-scale interviewing

We propose a two-phase, hybrid implementation, the Multi-Hybrid Agent Elicitation System for Tree-based Reasoning and Orchestration (*MHAESTRO*) which decomposes the process into two manageable parts. The Knowledge Engineering tool (*K-Eng*) produces a rule-based conversational survey framework encoded as a JSON decision tree that specifies prompts, branching conditions, required topics, and per-node adequacy settings. The Elicitation tool (*Elicitor*) then carries out the conversational survey by strictly traversing this deterministic structure while using LLMs for phrasing, summarising, and mapping responses to the schema. This separation helps with scale (many respondents follow the same plan), with quality (mandatory topics cannot be skipped), and with governance (every step is defined in advance and can be inspected). In short, the method behaves like a survey in coverage and repeatability, while still allowing natural language interaction.

1.4 Structure of the Paper

The paper is organised as follows. Section 2 reviews related work; Section 3 presents the hybrid architecture and governance controls; Sections 4 and 5 set out the methods and results of a formative case study; Section 6 discusses implications, including the consistency–fluidity trade-off and adjustments to reduce respondent burden while retaining structure and finally Section 7 concludes and outlines future work.

2 Literature Review

Knowledge elicitation remains constrained by the classic acquisition bottleneck: scarce expert time, high effort, and ad-hoc verification. Structured conversational surveys offer repeatable coverage and easier analysis, but can be rigid without a mechanism for context-sensitive probing. Modern dialogue systems and multi-agent orchestration promise natural interaction and role separation, yet LLM-led pipelines bring risks; hallucination, topic drift/user steering, privacy concerns, and opaque control. Prior survey-methods work and accessibility standards underline that validity (coverage, consistency, provenance, participant burden) is inseparable from governance. We therefore review: (i) structured surveying and adaptive instruments; (ii) dialogue/multi-agent designs as governed pipelines; (iii) LLM strengths and risks for

elicitation; and (iv) governance and EDI/accessibility principles. We end by identifying the gap: combining LLM adaptivity with deterministic, auditable control, precisely the role of our *K-Eng* → *Elicitor* architecture.

2.1 Structured surveying and adaptive instruments

Knowledge elicitation seeks to extract expert knowledge systematically but remains limited by the classic acquisition bottleneck: scarce expert time, high effort, and ad-hoc verification (Shadbolt and Smart 2015; Hoffman 1989). Within this space, structured conversational surveys and structured interviews provide repeatable coverage and analysis-ready outputs (Cooke 1994; Shadbolt and Smart 2015; Hoffman 1989). Their drawback is rigidity: fixed scripts can cap depth and restrict context-sensitive probing (Robinson 2023; Edwards and Holland 2013; Brayda and Boyce 2014). The survey-methods canon tackles reliability and traceability via schema-driven instruments, pretesting and cognitive interviewing, and explicit protocols for coverage and provenance (Tourangeau et al. 2000; Groves et al. 2009; Willis 2005; Schober and Conrad 1997; West et al. 2018). In parallel, computerised adaptive testing shows how item selection can adapt to responses while preserving measurement validity (Wainer 2000; van der Linden and Glas 2010). Our work carries these concerns, coverage, depth, and auditable traceability, into a governed, conversational setting.

2.2 Dialogue and multi-agent designs as governed pipelines

Task-oriented dialogue systems support schema-aware interaction, from slot-filling pipelines to end-to-end neural dialogue state tracking (Young et al. 2013; Wen et al. 2017). LLMs extend this by rephrasing, following up, and adapting to earlier answers. Recent studies suggest GPT-4–class models can draft follow-ups of comparable quality to human-authored ones in requirements elicitation (Shen et al. 2025; Siro et al. 2024). Moving from a single pipeline to multi-agent orchestration adds something needed for governance: role separation, message-level logs, and inspectable control points. The AutoGen (Wu et al. 2024) framework formalises role-specialised, message-passing agents that can be composed into auditable workflows (Commander/Writer/Critic patterns). Survey work now maps the multi-agent design space, including planning, tool use, and coordination failure modes that matter for pipeline control (Guo et al. 2024). Tooling is also maturing: AutoGen Studio (Dibia et al. 2024) exposes visual traces, costs, and tool-call histories that support debugging and auditability in governed elicitation. In short, dialogue agents provide adaptivity; multi-agent orchestrations provide roles, logs, and control.

2.3 LLM strengths and risks for elicitation

LLMs bring three advantages that directly support elicitation: (i) naturalistic interaction and rephrasing, which may aid comprehension, though effects on burden can vary with added clarification turns (Conrad and Schober 2000; Galesic and Bosnjak 2009); (ii) semantic transformation and summarisation to turn long answers into structured artefacts(Dagdelen et al. 2024; Jiao et al. 2023); and (iii) tool use to improve factuality and provide provenance (Lewis et al. 2020; Nakano et al. 2022; Menick et al.

2022). Set against this are documented risks: hallucination, topic drift/user steering, opacity of internal state, and privacy leakage (Ji et al. 2023; Carlini et al. 2021). Self-checking and verification methods (e.g., self-consistency, chain-of-verification) can reduce non-factual content but do not, by themselves, guarantee governed traversal or coverage (Manakul et al. 2023; Dhuliawala et al. 2024). Recent security and privacy surveys catalogue concrete attack surfaces (prompt injection, data exfiltration, membership inference) and defence patterns, underscoring the need for deterministic guardrails and provenance (Yao et al. 2024; Rathod et al. 2025; Shen et al. 2024; Rebedea et al. 2023). Early deployments of LLM-mediated conversational survey collection are feasible at small scale but highlight the importance of accuracy checks and participant-experience monitoring (Kaiyrbekov et al. 2025; Wuttke et al. 2025).

2.4 Governance and EDI/accessibility principles

Survey methods link validity to governance. Classic work stresses rules for coverage (reach required topics), consistency (comparable prompts/probes), and provenance (traceable answers and rationales), alongside participant burden as a determinant of break-off and candour (Tourangeau et al. 2000; Groves et al. 2009; Schaeffer and Dykema 2011; Galesic and Bosnjak 2009). In collaborative elicitation, change history and annotation have been used to strengthen provenance for verification and reuse (Tudorache et al. 2013).

Conversational interviewing shifts risks and benefits. Studies show interviewer effects and respondent steering as recognised threats to consistency, yet also document gains in accuracy for complex items when clarification is allowed (Schober and Conrad 1997; Conrad and Schober 2000; West et al. 2018). The Total Survey Error perspective frames these trade-offs across contact, measurement, and processing.

In AI-mediated pipelines, governance emphasises documentation, role separation, and auditability. Model reporting (e.g., Model Cards) and dataset documentation (e.g., Datasheets) are proposed to disclose scope, evaluation (including disaggregated results), and intended uses; end-to-end internal auditing frameworks address accountability through lifecycle artefacts (Mitchell et al. 2019; Gebru et al. 2021; Raji et al. 2020). Reviews of AI ethics guidelines highlight recurring principles, transparency, fairness, responsibility, privacy, while legal analyses caution that explanation rights are limited and context-dependent (Mittelstadt et al. 2016; Jobin et al. 2019; Wachter et al. 2017; Veale et al. 2018).

Accessibility is treated as part of validity as well as ethics. Prior guidance recommends plain language, adjustable pacing, and multimodal I/O compatible with assistive tech, with conformance to established criteria (e.g., WCAG 2.2 and W3C COGA) to reduce avoidable barriers (Lazar et al. 2015; W3C 2025, 2021).

2.5 Gap and contribution

Across these strands, the gap is to combine LLM-level adaptivity with deterministic, auditable control. Human-led interviewing offers depth but invites interviewer/agent effects; form-based instruments give consistency but lack probing; LLM pipelines are adaptive but probabilistic and hard to audit (Conrad and Schober 2000; West et al.

2018; Ji et al. 2023). Existing agent frameworks can enforce control flows, but they are general platforms rather than a governed end-to-end workflow for auditable knowledge elicitation (Wu et al. 2024; Guo et al. 2024). Our *K-Eng* → *Elicitor* architecture targets this gap by (i) compiling expert-validated policies into a versioned decision tree, (ii) executing a strictly traversed conversational survey with role-separated adequacy checks, producing a turn-by-turn logs and schema-valid outputs and (iii) answer the question: How can conversational knowledge elicitation achieve auditable control without undermining participant agency and accessibility? This positions conversational surveying as a governed pipeline with measurable validity (coverage, consistency, provenance, and participant burden) rather than an opaque chat.

3 The Proposed Hybrid Knowledge Flow Framework

This section formalises the hybrid framework implemented in *MHAESTRO* and explains how knowledge flows from expert input to auditable conversational survey outputs. The design goal is to achieve consistent coverage without losing the ability to probe adaptively, while leaving a clear audit trail. To meet this, the framework separates concerns into two phases linked by an explicit rule set.

Phase 1: Knowledge Engineering (K-Eng): A survey agent works with a domain expert to produce a structured, decision tree that specifies topics, branching conditions, and required depth of response. This turns a labour-intensive modelling step into a repeatable artefact that agents can execute deterministically.

Phase 2: Knowledge Elicitation (Elicitor): A coordinated set of agents, a survey agent, a steering/adequacy checker, and a traversal controller, conducts the session by following the tree. The steering agent blocks progression until the current node's requirements are met; the traversal agent advances along the correct branch and records justifications and metadata needed for audit.

Figure 1 summarises this two-phase architecture. The subsections that follow describe the phases and agents in detail, the prompt/context strategy, and the decision-tree schema that provides the deterministic backbone for execution.

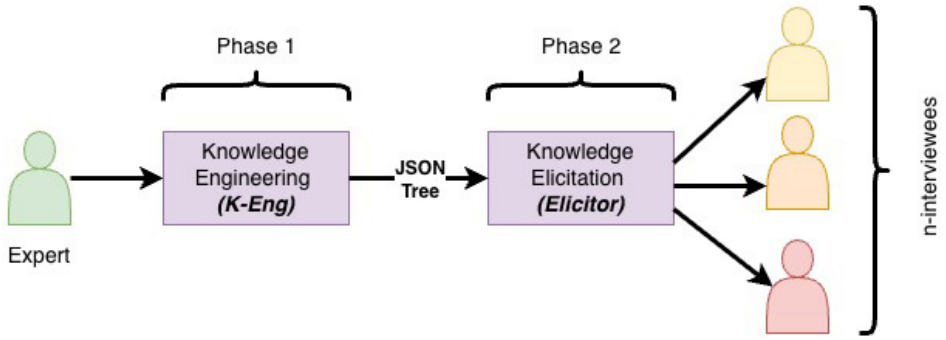


Fig. 1 Two-phased Hybrid Knowledge Flow Framework Architecture

3.1 Rationale for the Hybrid Approach

The architecture explicitly combines two historically divergent AI methodologies. The framework benefits from:

1. **Data-Driven Element:** The generative capacity of the LLM is used during *K-Eng* to rapidly synthesise and organise complex domain knowledge, avoiding the manual encoding of every single rule.
2. **Rule-Based Element:** The decision tree functions analogously to a rule-based inference mechanism, providing the deterministic control flow and repeatability necessary for reliable and scientifically rigorous structured conversational surveying. For instance, the system adheres to the logic: "if condition A holds, then ask question B".

This hybrid approach ensures that while the survey content is flexible enough to respond conversationally, the overall structure remains consistent and goal-oriented, minimising the risk of the survey questions veering off-track due to generative drift or hallucination.

3.2 The Knowledge Engineering Tool (*K-Eng*): Expert-Guided Tree (Codified Structure)

The *K-Eng* is critical, as the quality and completeness of the decision tree directly determines the efficacy and consistency of the execution phase.

3.2.1 Architecture and Operation

MHAESTRO aims to solve the issue of resource commitment by limiting the domain expert's required time to a focused conversational survey with the *K-Eng* tool, with an optional review of survey transcripts produced by the *Elicitor* tool. The *K-Eng* tool employs a single LLM survey agent (powered by GPT-4). This agent elicits comprehensive details regarding the necessary questions, underlying reasoning, and any domain-specific nuances required for the conversational survey framework.

3.2.2 Prompt Engineering and Context Management

All prompts governing the agents' behaviour are externalised and stored in a semicolon-delimited CSV file, allowing for easy configuration and adjustment. The system employs persona-based prompting techniques to define the roles of the agents. During piloting, we observed improved adherence when we emphasised key tokens (e.g., capitalisation or use of "**" to highlight context elements); given known sensitivity of LLMs to prompt formatting and context placement, we standardised this convention (Liu et al. 2024; Pezeshkpour and Hruschka 2024; Zhu et al. 2024).

3.2.3 The Decision Tree as Deterministic Backbone

We run the structured conversational survey with open-ended questions in the *Elicitor* using a pre-validated decision tree loaded from a JSON artefact produced in the

preceding analysis phase (Section 3.2.4). The tree provides the deterministic backbone for control flow: given the current node and the parsed category of the expert's answer, the next node is fixed by the tree's labelled edges. For clarity, we use "tree" as shorthand; the control structure may admit joins ($in-degree > 1$) and is therefore a directed acyclic graph (DAG), not a strict tree.

At run time, *Elicitor* begins at a root node that scopes the domain and presents the starting prompt defined in the decision tree. From any node, the system asks the node's question, the language model renders and clarifies that prompt, and the free-text answer is mapped to one of the node's predefined condition labels. Traversal itself is rule-based: the selected label determines the unique child to visit next. A node is treated as a leaf when no outgoing edge applies to the current answer category. Upon reaching a leaf, the system emits a structured summary and requests a brief confirmation or any further clarification from the participant.

The importance of the decision tree in our hybrid approach allows for reproducibility across sessions in a representation that is easy for experts to review (see Figure 3 as an example). Furthermore, it ensures that the free-text element of the LLM survey agent is restricted to ensure a natural flow to the question and conversational survey sequence, while still ensuring all pre-determined, vital questions and points are covered. To preserve determinism, the *Elicitor* consumes only JSON that passes schema and consistency checks before a session begins. We validate schema and graph consistency before each session.

3.2.4 Decision Tree Generation and Output

After each *K-Eng* conversational survey, which involves a single LLM agent conversationally extracting the following information from the expert in the following order: starting questions, mandatory questions, links between questions, and terminating questions, an *Analysis Agent* (GPT-4) is invoked once to read the transcript and synthesise a candidate decision tree in JSON. The process can be cascaded to be iterative across sessions: *update* (propose a revised tree), *validate* (schema and logical checks, including id/label stability), then *execute* (use the most recent validated tree in the *Elicitor* at the next session).

Each node carries a stable `id`, a `prompt`/`question` field, an `objective` field outlining the underlying reasoning of the question being asked, and an ordered `children` array. Each child is a transition object with a categorical `condition` label and a `target_id` pointing to the next node. Labels used at run time must match these `condition` values exactly; otherwise the *Elicitor* will raise a traversal error.

The *Analysis Agent* is given a strict JSON schema to adhere to, including conformity to specific label identifiers, in order to produce a valid tree. Within *MHAESTRO*'s hybrid pipeline, we set GPT-4's `response_format` to JSON to prevent markdown artifacts from contaminating component I/O and to simplify parsing, thereby reducing prompt-engineering burden, consistent with evidence that grammar-constrained decoding improves syntactic well-formedness for structured outputs, albeit with potential reasoning trade-offs (Park et al. 2024; Raspanti et al. 2025; Tam et al. 2024).

3.3 The Knowledge Elicitation Tool (*Elicitor*): Governed survey agent (Adequacy-Checked Flow)

Figure 2 presents the overall architecture of the *Elicitor* which utilises a multi-agent orchestration to conduct the structured conversational survey in real-time, leveraging the compartmentalisation of tasks to maximise consistency and maintainability. The architecture operates as a distributed yet unified procedure where specific agents handle questioning, steering, and traversal logic.

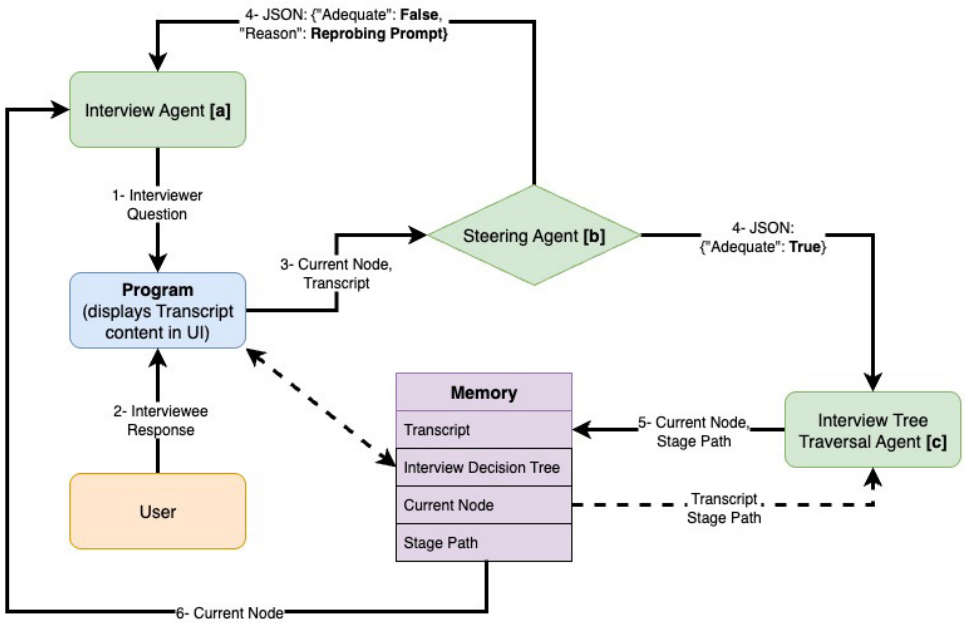


Fig. 2 Multi-Agent Orchestration Framework in *Elicitor*

3.3.1 Agent Orchestration and Compartmentalisation

The orchestration adheres strictly to the input decision tree, using its rules to guide the conversational survey flow. This approach minimises the likelihood of conversational survey content causing the conversation to stray off track, effectively reducing the impact of potential hallucinations on the overall structure. We chose GPT-4 after internal screening, consistent with comparative benchmarks showing stronger constraint-following (including JSON output format) relative to alternatives (Jiang et al. 2024).

Agents exchange critiques and decisions via a hidden JSON channel. This relay maintains the structural integrity of the dialogue and keeps roles compartmentalised;

the visible interaction remains stepwise and rule-bound. It functions like an interview panel or team with specialised roles within a governed pipeline.

3.3.2 Agent [a] The survey agent

Agent [a] serves as the main interface through which the participant interacts with the system. This agent is prompted to ask perceptive questions aligned precisely with the content of the current node in the decision tree, which is injected as a variable in its system prompt. The LLM's large knowledge base ensures that the generated questions are domain-pertinent. The agent's generated questions and the participant's responses are appended to the session state, generating the transcript used by subsequent agents.

3.3.3 Agent [b] The Steering Mechanism and Adequacy Check

The Steering Agent, Agent [b], is fundamental to the system's structural rigour. Its primary role is to act as an adequacy checker, ensuring that the participant's response contains sufficient information to satisfy the requirements outlined in the current decision tree node before progression is permitted.

Mechanism of Control: Agent [b] receives the current node and the full survey transcript. It analyses the content and returns a structured JSON object containing the boolean attribute adequate. If adequate is determined to be False, the JSON object also includes a string attribute detailing the reason for the inadequacy. This critique is then appended as an invisible system prompt (not visible to the user) to Agent [a]'s input for the next cycle. This system-prompt augmentation compels Agent [a] to immediately re-probe until adequacy is met and, because Agent [b] is the last control point before the next question, it also reduces the risk of the user steering the conversation away from required topics. This mechanism helps prevent omission of essential details and approximates the diligent follow-up of a skilled survey agent, while systematically improving the quality of follow-up questions via targeted critiques. Prior work underscores that probing is essential yet resource-intensive, Cognitive Task Analysis (CTA) protocols (e.g., the critical decision method) rely on intensive probe questions, and conversational clarification improves interpretation but substantially lengthens surveys, while human survey agents often struggle to craft high-quality follow-ups under real-time cognitive load (Shen et al. 2025; Hoffman 1989; Conrad and Schober 2000). In this study, adequacy was operationalised as a binary decision: a turn was adequate when the participant's answer contained (a) a topical stance and (b) one supporting reason; otherwise Agent [b] triggered an immediate re-probe.

3.3.4 Agent [c] The Tree Traversal Logic

The Tree Traversal Agent, Agent [c], is responsible for moving the conversational survey through the decision tree structure once Agent [b] confirms that the requirements of the current node have been met.

Deterministic Flow: If the current node is a leaf node, the conversational survey concludes. If the node has only one child, the tree structure is updated to move directly to the next node. If the node has two children, indicating a conditional branch, Agent [c] is called. Agent [c] is provided with the conversational survey transcript and

the path taken so far, advancing along the labelled edge to the uniquely determined child node. This decision is outputted as a JSON object, complete with a justification for the choice, which is currently used for logging and transparency purposes. This strict, conditional traversal ensures the underlying process adheres to the rule-based logic encoded in *K-Eng*.

3.4 Governance and audit controls

Our implementation instantiates the quality and governance concerns outlined in Section 2.4 as follows: deterministic *coverage* via decision-tree traversal with per-node adequacy thresholds; *transparency* via per-turn logs that record timestamps, node IDs, agent role, model/version, and prompt template ID; concise *rationales* captured for each supporting agent's decision, especially in the vital re-probing and tree traversal stages; *provenance* through recording the decision-tree commit hash and prompt template hash; drift and user-steering *control* through checks that compare the current turn to required slots/topics and issue corrective prompts; and *human oversight* is also available in all conversational surveys and in future iterations, will be a vital step in the *K-Eng* for review and synthesis of the decision tree. With logging enabled, these controls make runs inspectable where an auditor can trace why a question was asked, which policy version applied, and which agent executed the step, addressing the opacity typical of purely generative pipelines. Table 1 summarises how we translate the governance requirements into concrete controls, and flags any items not collected in this study.

Governance requirement	Implementation in our system
Coverage and consistency	Deterministic decision-graph (DAG) traversal with per-node adequacy thresholds.
Transparency and auditability	JSON logs per turn (timestamp, node ID, agent, model/version, prompt template hash).
"Why" rationales	Supporting agents must justify why a decision has been made.
Adequacy/Uncertainty	`adequacy` field (True/False) with optional justification when evaluating whether an participant's answer is sufficient.
Provenance	Decision-tree commit hash and prompt template hash recorded per interaction. At each stage, the logs also show all messages and interactions available to each agent.
User steering control	On-turn checks against required topics applicable to the current node; corrective prompts when off-path.
Human oversight	Transcripts can be downloaded and monitored.
Data protection	Consent, pseudonymisation, and restricted storage on approved institutional drives. Information elicited from participants should not be used for training any models.

Table 1 Operationalising the governance concerns from Section 2.4 in our implementation.

4 Experiment Design and Methodology

This section describes the study context, participants and procedure, the outcome measures we used to judge performance, and how control and governance were operationalised in the prototype. The goal was to assess both extraction efficacy and user experience in a small formative evaluation, while keeping the procedure auditable and repeatable.

4.1 Context and Application Domain

We deployed the system in an educational, EDI-relevant setting centred on a short card-game activity about the meaning of names and cultural awareness. After playing, participants completed a structured conversational survey with open-ended questions using the *Elicitor*. The conversational survey framework was pre-generated by the *K-Eng* as a JSON decision tree. While the domain allowed us to study high-level conceptual knowledge (culture, inclusion, social understanding), the core technical objective was testing whether a deterministic survey plan could be followed transparently in practice and whether the governance controls introduced a fidelity-fluidity trade-off.

4.2 Participants and Procedure

Eight adult volunteers (students and staff from across the university) took part on campus. Following the card game, each participant used a laptop or phone to complete the *Elicitor* session. All sessions used the same pre-created decision graph so that the structure of questions and branching conditions was identical across participants. For readability we refer to this structure as a "tree", though joins make it a rooted directed acyclic graph (DAG). Each session started at a root node, the system posed the node's question, mapped the free-text answer to one of the node's predefined condition labels, and then followed the uniquely determined child edge. At a terminal leaf, the system displayed a brief structured summary and asked the participant to confirm that their points had been captured.

4.3 Measures

We defined two outcome families to capture both success of extraction and quality of experience. All rating items used a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree), presented with a short on-screen instruction that remained visible throughout the session.

4.3.1 Extraction Fidelity (Summary Accuracy)

To measure the system's fidelity, a summariser agent generated a short end-of-session summary from the transcript. Participants then indicated whether this summary *accurately reflected* what they had provided. Participants then rated the accuracy of this summary on a 5-point scale. This metric does not measure the participants' valuation

of the content's topic, but rather the system's technical ability to capture and reproduce that input faithfully. Agreement on this check is taken as a direct indicator of extraction fidelity (i.e., low omission/addition risk).

4.3.2 User Experience (subjective quality)

We captured perceptions of interface usability, conversational flow/coherence, pressure or intrusiveness (reported repetitiveness), visual appeal, and a comparison to the experience of being surveyed by a human. We also included a single human-equivalence judgement ("did this feel like interacting with a person?") and recorded simple associations between that judgement and the UX facets for exploratory insight; the analysis itself appears in Section 5.

4.3.3 Coding and Scoring

All Likert items used a 1–5 agree scale (1 = Strongly Disagree, 5 = Strongly Agree). To keep directions aligned so that higher always indicates a better user experience, the repetitiveness/intrusiveness item was reverse-coded as $x' = 6 - x$ and reported as *Non-intrusiveness*. Construct summaries report *item-level* statistics unless stated otherwise. For the EDI/Content block (5 items $\times$ 8 participants; $N_{\text{obs}} = 40$) we report the item-level mean (4.2) and median (5.0). Where participant-level means are used, we state this explicitly.

4.4 Operationalising Control and Governance

Control was enforced by a Steering Agent that evaluated each turn against the current node's objective and returned a structured adequacy flag. Progression was deterministically blocked until adequacy was met; only then did the Tree-Traversal component advance to the next node. Completion was defined as reaching a terminal leaf.

4.5 Binding to governed artefacts

All sessions were executed on the pre-validated decision tree from Section 3.2 (tree commit with the associated prompt-template bundle and adequacy policy set; these identifiers were fixed in the study protocol and checked at initialisation to ensure a single, identical configuration across sessions.

To support audit and accountability, the prototype includes a per-turn logging schema for timestamps, node IDs, agent role, model/version, and prompt-template identifiers. Node identifiers and timestamps were not included in the analytics because they were not pertinent to the research questions (which concern content-level outputs and user-reported experience); their omission does not affect the results or interpretations. Supporting agents produced concise "why" justifications for key decisions. Data handling followed consent and pseudonymisation practices, with restricted storage and no reuse of elicited information for model training. Table 1 summarises how governance requirements (coverage/consistency, transparency, provenance, on-turn user steering, human oversight, and data protection) were instantiated in the prototype.

5 Results and Analysis

This section reports what the study log shows about coverage, experience, and the central design trade-off. We keep the analysis descriptive given the small sample, using medians, non-parametric checks, and bootstrap intervals to support transparency. All numbers are computed directly from the attached per-turn log.

5.1 Data preparation and scope

The log contains two categories per participant: five Survey Evaluation items (each paired with a system summary answer) and five UI Feedback items, all on a 1–5 Likert scale (1=Strongly Disagree, 5=Strongly Agree). To keep directions consistent, the repetitiveness/intrusiveness item is reverse-coded as $x' = 6 - x$ and reported as non-intrusiveness. With N=8 participants, we treat analyses as descriptive and report means, medians, standard deviations, Spearman correlations, and bootstrap confidence intervals where helpful.

5.2 System-Level Verification and Traceability Audit

A core claim of *MHAESTRO*'s framework is its ability to ensure consistent coverage and mitigate user steering by "strictly traversing" the pre-defined decision tree. To create the decision tree seen in Figure 3, the following five key questions were passed to the agent in the 'expert' interview when using the *K-Eng* tool (Section 3.2) and were then used again in the experiment to confirm the success of the hybrid system's implementation by having the summary agent explore these same key points (Section 4.3.1):

1. In a social situation, have you ever wondered or wanted to find out the meaning of somebody's name?
2. If yes, what triggered you to ask that question
3. Do you believe it's important to know the meaning of other people's names?
4. Does an understanding of name etymology make you a more diverse thinker and EDI aware?
5. Does name etymology knowledge increase your curiosity about different cultures and does this knowledge empower you within a social circle?

By filtering the data for the survey evaluation category, the exact path for any participant can be reconstructed from the decision tree matching per-turn question texts in the data to nodes in Figure 3. The data logs provide an immutable, turn-by-turn trace to verify this claim.

This demonstrates the *Elicitor*'s successful adherence to the *K-Eng* phase's governance structure, confirming that progression is controlled by the system's internal logic, specifically the internal structured JSON control, rather than by off-topic user turns.

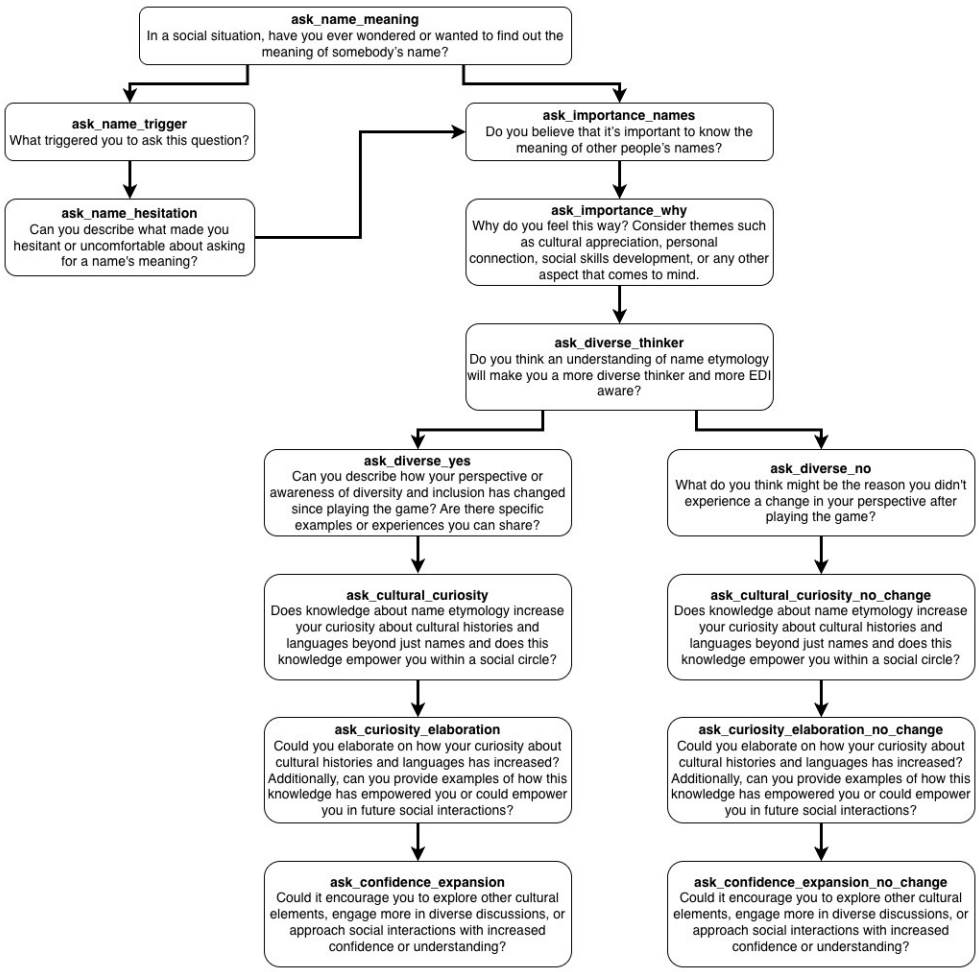


Fig. 3 Decision tree produced through the experiment's *K-Eng*. Terminology note: "tree" is used informally; the underlying structure is a directed acyclic decision graph with occasional joins ($in\text{-}degree > 1$).

5.3 Construct-level outcomes

The extraction fidelity items (five per participant; 40 observations) , which measure the perceived accuracy of the system's summary, are strongly positive overall: M=4.20, $Median$=4.50, SD=0.88. This indicates a high degree of technical success in capturing user input.

In contrast, the Aligned UX facets (ease, visual appeal, coherence, non-intrusiveness; 32 observations) are moderate: M=3.22, $Median$=4.00, SD=1.26. Perceived human-equivalence (as effective as a human-led survey) sits below neutral: M=2.75, $Median$=2.50, SD=1.28.

This reveals a key tension: while six of eight participants averaged ≥ 4 for extraction fidelity (technical success), only one of eight averaged ≥ 4 across aligned UX (human-centric success). See Table 2 for a compact summary.

5.4 Internal consistency

The five extraction fidelity items form a reliable set (Cronbach's α=0.89), justifying their use as an aggregate fidelity score. The four aligned UX facets do not behave as a single scale (α=0.54), so we interpret UX item-wise (ease, visual, coherence, non-intrusiveness) throughout the section.

5.5 Item-level patterns

The extraction fidelity signals are uniformly positive (overall 75% agreement, ratings ≥ 4 on a 1–5 scale), with high mean scores for summary accuracy across all five items (e.g., importance M=4.38, diverse-thinker M=4.25).

The UX facets, however, reveal a sophisticated tension. The system's surface-level usability was high: Ease of use is strongly positive (M=4.13, SD=0.99, 87.5% agreement). In contrast, the metric related to the system's core governance logic was very low. On the original scale, participants agreed the survey felt repetitive/intrusive (M=4.25, SD=1.04). Expressed as the aligned metric Non-intrusiveness, this score is M=1.75 (SD=1.04).

This split is the paper's central finding that participants found the interface easy to use, but the governed interaction intrusive and repetitive. Visual appeal and coherence were moderate (both M=3.50). These patterns are visible in Table 2 and motivate the trade-off analysis in Section 5.8.

Table 2 Summary of constructs and key items (N=8; 1–5 Likert).

Construct / Item	**Mean**	**Median**	**SD**
Extraction Fidelity (Summary Accuracy) (40 obs)	4.20	4.50	0.88
UX (aligned) (ease, visual, coherence, non-intrusiveness; 32 obs)	3.22	4.00	1.26
Human-equivalence (single item; 8 obs)	2.75	2.50	1.28
Ease of use	4.13	4.00	0.99
Visual appeal	3.50	4.00	0.76
Coherence	3.50	3.50	0.93
Repetitive/intrusive (raw)	4.25	4.50	1.04
Non-intrusiveness (aligned)	1.75	1.50	1.04

5.6 Associations with perceived human-equivalence

Exploratory participant-level correlations link perceived human-equivalence to UX facets. Non-intrusiveness shows the strongest directional association (Pearson r=0.59; Spearman ρ=0.33), followed by coherence (r=0.48; ρ=0.37), ease (r=0.37; ρ=0.38), and visual appeal (r=0.29; ρ=0.26). As expected for N=8, bootstrapped 95% confidence intervals for Pearson r are wide (e.g., non-intrusiveness [−0.65, 0.96]; coherence [−0.53, 0.90]), but the ordering is stable across Pearson and Spearman. The directional reading is straightforward: the less repetitive/intrusive and the more coherent the experience, the more human-like it feels. Table 3 reports the full set.

Table 3 Exploratory correlations with human-equivalence (participant-level, N=8). Pearson r and Spearman ρ; 95% CI for Pearson via bootstrap (10,000 resamples).

Facet	**Pearson r**	**Spearman ρ**	**95% CI (Pearson)**
Non-intrusiveness	0.59	0.33	[−0.65, 0.96]
Coherence	0.48	0.37	[−0.53, 0.90]
Ease of use	0.37	0.38	[−0.28, 0.79]
Visual appeal	0.29	0.26	[−0.36, 0.86]

5.7 Qualitative signals from per-turn summaries

The system's *Summary Answer* fields for Survey Evaluation (40 entries) repeatedly mention cultural and social themes: *culture*=24, *diverse*=11, *inclusion*=7, *social*=12, with sparse mentions of $confidence$=1. This mirrors the quantitative pattern: the *why* (cultural understanding and inclusion) resonates, while comfort and naturalness are more fragile.

5.8 Quantifying the Fidelity-Fluidity Trade-Off

To express this central tension, we propose a fidelity-fluidity gap as the difference between the mean extraction fidelity and the mean Non-intrusiveness: $M_{\text{Fidelity}} - M_{\text{Fluidity}} = 4.20 - 1.75 = 2.45$

This 2.45-point gap is not merely correlational; it is the quantitative, empirical measure of the system's core socio-technical trade-off. The system's high fidelity (4.20) is achieved because of its governance mechanism (the Steering Agent [b]), and its low perceived fluidity (1.75) is the direct human-centric cost of that same mechanism.

5.9 Takeaways

In short, the governed process achieved full coverage and high extraction fidelity; the experience also carried significant interaction strain, captured by high repetitiveness (low non-intrusiveness) and lower human-equivalence. These results provide concrete

baseline metrics for the next iteration (e.g., raising non-intrusiveness and human-equivalence) while preserving coverage completeness. The Discussion picks up this trade-off, relates it to governance aims, and outlines control agent to reduce interaction pressure without losing traceability.

6 Discussion

The findings in Section 5 show a clear, quantitative trade-off. This paper's central finding is not simply that a trade-off exists, but that the pattern is consistent with a causal interpretation linked to governance-by-design. The perceived repetitiveness (M=4.25) and low non-intrusiveness (M=1.75) are the immediate, human-facing consequences of the governance-by-design implemented by Agent [b], the steering mechanism and adequacy check (described in Section 3.3.3). This agent's function to block progression and compel Agent [a] to immediately re-probe is what ensures high fidelity and what creates the user experience of being pressured and badgered. This demonstrates a new form of AI-driven participant burden, extending the classic survey-methods concept (Tourangeau et al. 2000; Galesic and Bosnjak 2009) into governed human-AI interaction.

6.1 Consistency at scale versus participant agency

MHAESTRO aims to secure consistent knowledge coverage through governance mechanisms designed in the *K-Eng* phase and executed by the *Elicitor* (Figure 3). This trade-off is not abstract; the pattern is consistent with a causal interpretation linked to the system's architecture. The perceived repetitiveness (M=4.25) and low non-intrusiveness (M=1.75) are the immediate, human-facing consequences of the governance-by-design implemented by Agent [b] the steering mechanism and adequacy check (Section 3.3.3). This agent's function to block progression and force Agent [a] to immediately re-probe is what ensures high fidelity and what creates the user experience of being pressured and badgered. The log confirms coverage completeness across all five survey items for every participant, and the correlations in Table 3 indicate that the facets most closely tied to a more human-like feel are non-intrusiveness and coherence. In plain terms, the very controls that ensure coverage also risk making the exchange feel rigid. This gives a concrete design direction which is to introduce an on-turn agent without losing traceability. Practical options include a visible skip/park control for a question, a brief explain-why hint that makes the governance goal of the current turn legible to the participant, and a soft cap on repeated adequacy checks so that the system refrains from pressing beyond a reasonable point. These are small, auditable changes that target the two weakest outcomes directly.

This tension between technical success and social experience is quantified in Table 4, which contrasts key metrics for technical performance against those for perceived interaction quality. This data reveals that the problem is not a poor interface (Ease of use is high) but a punishing interaction logic (Non-intrusiveness is low), further implicating the governance mechanisms of Agent [b] as a plausible contributor to the negative social outcome.

Table 4 The fidelity-fluidity trade-off quantified: A contrast between technical performance (Fidelity, Ease of Use) and perceived interaction quality (Non-intrusiveness, Repetitiveness).

Construct	Metric	Mean score (1–5)	Interpretation
Technical Fidelity	Extraction Fidelity (Summary Accuracy)	4.20	High Success: The system works at its technical goal (capturing input).
Social Fluidity	Non-intrusiveness (aligned)	1.75	High Failure: The experience of this governance is felt as highly intrusive.
Usability (Interface)	Ease of use	4.13	High Success: The UI is clear and easy to use.
Usability (Interaction)	Repetitive/intrusive (raw)	4.25	High Failure: The governed interaction is perceived as repetitive.

6.2 Governance-by-design and auditable traces

For a socio-technical reading, the most important asset in the dataset is the per-turn trace. Each survey item is paired with a concise system summary, which provides an inspectable record of what the tool registered as the participant's answer. This supports accountability claims without relying on opaque internal states. At the same time, the current sheet records question text but not node identifiers, timestamps, or prompt-template versions. As a result, Results can verify coverage rather than exact sequence. If sequence and versioning must be auditable, the node ID, timestamp, and the prompt-template version per turn can be used to compute a short cryptographic hash of the emitted summary.

6.3 Equity, accessibility, and validity

The qualitative signals in the summaries emphasise cultural understanding and social connection, which supports the EDI intent of the task. For broader use, inclusion must be supported in the interaction layer, not only in the topic choice. General best practices, such as adjustable pacing, concise on-screen text, and optional audio read-out, can help users with different cognitive and language needs engage without fatigue.

This study's central finding, the interaction strain captured by the high repetitive/intrusive score (Raw M=4.25), is therefore not just a user experience problem, but a critical accessibility and equity issue. The avoidable strain caused by the system's persistent adequacy checks (Section 3.3.3) can be expected to disproportionately impact users with high cognitive fatigue, anxiety, or neurodivergent profiles who may find the unyielding, machine-driven probing stressful.

These are not only ethical choices; they also protect validity by reducing avoidable strain that can distort responses while coverage is enforced. This reframes the design

implications discussed elsewhere (such as a soft cap on repeated adequacy checks and a visible skip/park control) as essential accessibility features, necessary to protect validity by ensuring the system's governance does not distort responses from the very groups it aims to include.

6.4 Reading the correlations responsibly

With N=8, associations are exploratory. Reporting both Pearson and Spearman, plus bootstrap intervals, was a deliberate choice to avoid overstating precision. Even so, the ordering is informative: reducing repetitiveness and improving coherence are the most promising levers for raising perceived human-equivalence. This is a practical recipe for iteration: target those levers first and re-measure against the same metrics reported in Table 3.

6.5 Design implications and baseline-to-beat

The study provides a testable baseline for the next release. A reasonable goal is to increase aligned non-intrusiveness while keeping extraction fidelity at or above its current mean. The skip/park control, brief why-hints, and a limit on repeated adequacy checks can be implemented without changing the governed structure. After deployment, the same analytics used in Section 5 can be re-run to show whether the gap narrows and whether perceived human-equivalence improves.

6.6 Safeguarding and operational assurance

The per-turn summaries also create an opportunity for live safeguarding. A lightweight monitor could scan each new summary for signs of distress or risk and either surface a support message or pause progression for human review. If implemented, such interrupts should be recorded in the same log with an explicit event type so that the audit trail remains continuous and the intervention itself is accountable.

6.7 Limitations and next steps

The sample is small and confined to one EDI-themed task, so the quantitative results are descriptive. The absence of node identifiers and timestamps limits sequence auditing in this version of the log. These are tractable constraints. The natural next study is comparison between the current governed flow and a version with on-turn control agents, using non-intrusiveness and human-equivalence as primary outcomes and coverage completeness as a non-negotiable constraint. A further step is to compare fixed adequacy thresholds with simple respondent-state models that relax probing for slower or repeatedly pressed participants, under a fixed coverage constraint. Qualitative coding of the per-turn summaries can also move beyond keyword counts to examine whether control agent change the substance of what is recorded, not only the feel of the exchange. Qualitative coding of the per-turn summaries can also move beyond keyword counts to examine whether control agent change the substance of what is recorded, not only the feel of the exchange.

Taken together, the results show that governed conversational surveying can deliver consistent coverage and high extraction fidelity, but that the experience must be tuned to protect participant. The discussion above sets out concrete, auditable changes that speak directly to the weakest outcomes while preserving the governance aims. The contribution is therefore both technical and social: a workable path for making governed conversational surveying accountable at scale without making it uncomfortable to take part in.

7 Conclusion

This paper introduced a governed, two-phase approach to conversational surveying that couples LLM-level phrasing with deterministic control (*MHAESTRO*). By compiling expert-validated policies into a versioned decision tree (*K-Eng*) and executing them with adequacy-checked traversal (*Elicitor*), the method delivers consistent coverage, auditability, and repeatable runs at survey scale. In a formative study ($N = 8$), we demonstrated high extraction fidelity (M=4.20) with complete coverage, confirming the system's technical robustness. However, this study also empirically identified and quantified the human-centric cost: a fidelity-fluidity gap of 2.45 Likert points. This gap makes the central socio-technical design tension explicit, framing it not as a simple UX flaw but as a core ethical and accessibility challenge for governed AI.

The contribution has three core strengths. First, separating policy from execution clarifies responsibility and makes the system easy to inspect. Second, strict traversal driven by internal structured-JSON control curbs drift and user steering while keeping a natural language exchange. Third, per-turn summaries act as practical audit artefacts for assurance, reuse, and operational monitoring, assets that are rarely combined in prior LLM pipelines for surveying.

At the same time, the experience showed interaction strain (low aligned non-intrusiveness and reduced perceived human-equivalence). We proposed small, auditable additions to improve agency without giving up governed coverage: a visible *skip/park* option for a question, short "why this turn?" hints before re-probes, and a soft cap on repeated adequacy checks. The same log stream also supports a lightweight safeguarding component (Agent [d]) that can pause or flag risk, with each intervention recorded in the audit trail.

Limitations are recognised: a small sample, a single domain, and logs that, in this release, omit node identifiers and timestamps. These are straightforward to address. The next step is an A/B evaluation that compares the current governed flow against the agency-enhanced variant, holding coverage as a fixed constraint and tracking non-intrusiveness, human-equivalence, and time/interaction measures. Extending the log schema with node IDs, timestamps, prompt/model versions, and a short hash of each summary will enable direct sequence and provenance auditing.

In sum, this work offers a practical route to accountable, inclusive conversational surveying: keep a clear line of control and traceability while making agency visible and usable on each turn. This work proves that a *governed* AI is a *felt* AI, and the *feeling* is often one of uncomfortable control. The next challenge for the field is therefore not just to build auditable systems, but to design them in a way that reconciles governance

with human dignity, agency, and accessibility. The baseline metrics and audit trail reported here provide a foundation for this critical next step.

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Declarations

- **Ethics approval and consent to participate:** Ethical approval for this study was obtained through Loughborough University's online ethics application system (LEON). All participants received an information sheet outlining the study's purpose, procedures, and their rights. Informed consent was obtained from all participants prior to feedback collection.
- **Data availability:** The datasets generated and/or analysed during the current study are available in the Loughborough University Research Repository at https://repository.lboro.ac.uk/. Each dataset is assigned a DOI to ensure persistent and citable access. Access to certain datasets may be subject to restrictions in accordance with ethical considerations requirements. For further information or to request access to restricted data, please contact the corresponding author.
- **Competing interests:** The authors have no competing interests to declare that are relevant to the content of this article.

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